

When FASHION Meets AI



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AI programs are becoming an efficient assistant for human stylists. These can handle more attributes, process large amounts of data faster and learn users' styles more accurately

Among all the domains of human creativity, human fashion behaviour modelling is one of the biggest challenges given its random nature due to erratic irrationality, individual uniqueness, craziness in selection and adoption, cultural dependence, and individual and cultural distinctiveness. According to a survey, 93 per cent senior business executives feel that selection and promotions of employees are affected by their fashion and clothing choices. To monitor and track the fashion sense, significant time and effort is required. Moreover, employing fashion experts and stylists is cost-intensive, and competent experts and professionals are difficult to get.

Artificial intelligence (AI) artist program AARON, created in 1937 by artist Harold Cohen, was the first profound connection between AI and the human creativity.

There are many benefits of using AI programs for fashion designing. AI fashion program can process large amounts of data faster, learn users' style more accurately, memorise and process users' feedback, store descriptions of users' items, and help users become more organised and efficient.

Evolution of AI

AI is the science and engineering of making intelligent machines, more so intelligent computer programs. In layman's language, if a computer performs a function that is deemed to be intelligent

for humans, we can say the computer has intelligence—an amalgamation of knowledge and reasoning power.

AI has come a long way from the demo of the first running AI program, the Logic Theorist (LT) in 1956, to creation of ELIZA (an interactive program capable of participating in a discussion in English on any subject) in 1965, to the 1990s' key advances in machine learning, intelligent tutoring, case-based reasoning, multi-agent planning, scheduling, uncertain reasoning, data mining, natural language understanding and translation, vision, virtual reality, games, etc.

1997 saw The Deep Blue—an IBM super-computer that beat the world chess champion Garry Kasparov. By the late 1990s, web crawlers and other AI-based information extraction programs became indispensable in widespread use of the worldwide web.

Convergence of AI and fashion

Applying AI in the field of human creativity has been one of the greatest challenges for software engineers. Fashion changes with culture and time. For instance, Asian fashion style is quite different from European. A complete styling task requires a large number of attributes, including event, clothes, shoes, accessories, makeup and hairstyling. A stylist is also very personal.

AI stylist programs consider three major components:

1. Visual garment representation—a garment database that stores garment items by categories
2. Computational styling, which process-

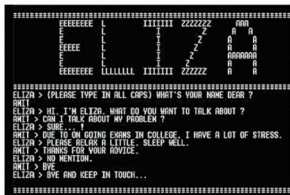


Fig. 1: ELIZA—an interactive program based on a very basic level of AI (Source: www.scaruffi.com)



Fig. 2: How fashion evolved through decades (Source: <https://fashionthroughthedecades.wordpress.com>)